$P(S_1 | S_1, a_1) = 0.8$ $\quad P(S_2 | S_1, a_1) = 0.2$

$P(S_1 | S_1, a_2) = 0.3$ $\quad P(S_2 | S_1, a_2) = 0.7$

$P(S_1 | S_2, a_1) = 0.6$ $\quad P(S_2 | S_2, a_1) = 0.4$

$P(S_1 | S_2, a_2) = 0.1$ $\quad P(S_2 | S_2, a_2) = 0.9$

$$P^{a_1} = \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix} \qquad P^{a_2} = \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix}$$

$$R^{a_1} = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix} \qquad R^{a_2} = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$$

i = row

j = col

$R^a_{ij} = R(S_i, a, S_j) \Rightarrow R(S_1, a_1, S_1) = 2$

$\pi(a|s)$: $\quad \pi(a_1 | S_1) = 0.7 \quad \pi(a_2 | S_1) = 0.3$

$\pi(a_1 | S_2) = 0.4 \quad \pi(a_2 | S_2) = 0.6$

$\gamma = 0.9$

a
1
2

s' r
1
2

1.a)

$$V^{\pi}(s_1) = \sum_a \pi(a|s_1) \sum_{s',r} p(s', r|s, a)\left[r + \gamma V^{\pi}(s')\right]$$

$$V^{\pi}(s_1) = 0.7 \times \left[0.8 \times \left(r + 0.9\, V^{\pi}(s_1)\right) + 0.2 \times \left(r + 0.9\, V^{\pi}(s_2)\right)\right]$$
$$+ 0.3 \times \left[0.3 \times \left(r + 0.9\, V^{\pi}(s_2)\right) + 0.7 \times \left(r + 0.9\, V^{\pi}(s_2)\right)\right]$$

$$= 0.7 \times \left[0.8 \times \left(2 + 0.9\, V^{\pi}(s_1)\right) + 0.2 \times \left(5 + 0.9\, V^{\pi}(s_2)\right)\right]$$
$$+ 0.3 \times \left[0.3 \times \left(1 + 0.9\, V^{\pi}(s_1)\right) + 0.7 \times \left(3 + 0.9\, V^{\pi}(s_2)\right)\right]$$

$$V^{\pi}(s_2) = 0.4 \times \left[0.6 \times \left(4 + 0.9\, V^{\pi}(s_1)\right) + 0.4 \times \left(1 + 0.9\, V^{\pi}(s_2)\right)\right]$$
$$0.6 \times \left[0.1 \times \left(0 + 0.9\, V^{\pi}(s_1)\right) + 0.9 \times \left(2 + 0.9\, V^{\pi}(s_2)\right)\right]$$

b)